Risk or Refuge? Bitcoin, S&P 500, and Bonds

This analysis was conducted in November 2025 and should be interpreted as an empirical observation rather than investment advice. The purpose is to explore how Bitcoin behaved alongside traditional assets during the selected period and whether it contributed to portfolio diversification or risk reduction. The findings are based on historical data and may not predict future performance. Investors should conduct their own research and consider their risk tolerance before making any investment decisions.

Executive Summary

This project investigates whether Bitcoin can improve diversified portfolio performance or whether it remains mainly a speculative risk asset. Using data from 2016 to 2025, we compare Bitcoin with the S&P 500, long-term U.S. Treasury bonds, and the VIX volatility index.

The analysis shows that Bitcoin remains highly volatile, but it has relatively low correlation with both equities and bonds. While it does not behave as a pure safe haven during market stress, it appears to offer diversification value when used in small allocations.

Portfolio backtesting across three frameworks — the 60/40 Portfolio, the All Weather Portfolio, and the Cockroach Portfolio — shows that modest Bitcoin allocations improved both annualized returns and risk-adjusted performance. The strongest results were achieved with Bitcoin allocations of approximately 4% to 9%, especially when funded from the bond component.

Overall, the findings suggest that Bitcoin should not replace traditional safe assets, but it may serve as a strategic complement in modern portfolio construction.

Introduction

Traditional portfolio construction has long relied on the idea that equities provide growth while bonds provide stability. However, in the post-Covid market environment, this relationship has weakened as stock-bond correlations increased and bond allocations became less protective.

At the same time, Bitcoin has become more relevant in portfolio discussions. Although it is often viewed as a speculative and volatile asset, its relatively low correlation with traditional assets raises an important question: can Bitcoin improve diversification and portfolio efficiency?

This project examines the relationship between Bitcoin, the S&P 500, long-term U.S. Treasury bonds, and the VIX volatility index from 2016 to 2025. It then tests whether adding Bitcoin to three portfolio frameworks — 60/40, All Weather, and Cockroach — improves performance.

The goal is not to present Bitcoin as a guaranteed safe haven, but to assess whether a small, carefully calibrated allocation can improve long-term portfolio outcomes.